

Outline Draft

Loveless, Nix, Stokes, Abernathy's

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1 Introduction

Alzheimer's Disease (AD) is characterized as a complex neurodegenerative disease affecting approximately more than 55 million people globally [10]. Symptoms are caused by the degeneration of the central nervous system, and are portrayed by progressive memory loss, difficulty with language, drastic mood swings, among other effects. It is a disease that causes an extensive burden to those suffering from the disease through disorientation, to relatives in the form of mental taxation, as well as economically; projecting an estimated global economic burden of 16.9 trillion dollars by 2050 [12]. These factors are but a few reasons as to the importance of research in this area, where many scientists have worked to understand the underlying mechanisms of AD. However, much is still unknown in how AD is caused, as well as what biological processes occur in the brain, making the disease difficult to come to a decisive conclusion.

Many studies have been conducted in regard to understanding Alzheimer's Disease. With this, there are a few main hypotheses that are generally considered to be factors in the causation and progression of AD [16]. The so-called *Amyloid Cascade Hypothesis* is among the most extensively studied in the literature, where it is proposed that a misfolding of β -amyloid ($A\beta$) protein form senile plaques. These plaques are hypothesized to be the main driving force in the intracellular deposition of misfolded tau proteins in neurofibrillary tangles (NFTs) and memory loss in the progression of Alzheimer's Disease [7]. Growing evidence however, shows that this hypothesis is suggested to be but a small factor in the larger picture of the underlying operative mechanisms of the disease [8]. This has led researchers to branch into other pathways concerning AD, looking at the interactions between $A\beta$ and tau in an attempt to further understand the pathogenesis of these two factors [15]. Some studies have also observed the role of Calcium in this process, further suggesting a more complex process that is occurring biologically. This occurs through a feedback interaction between an upregulation of Ca^{2+} and the amyloidogenic pathway, remodeling the Ca^{2+} signaling pathway [4].

The various complex interactions in AD have shown to be an obstacle to yet overcome when modeling the behavior of the disease. Numerous different studies have in turn given an attempt at creating mathematical models observing different aspects of the disease in order to observe the dynamics in occurrence. This has led to many models being introduced, each aiming to understand various components in Alzheimer's Disease through different methodologies. Compartments concerning the interaction between $A\beta$ and tau are studied, observing the suspected effects on neurodegeneration and cognitive decline through a mechanistic causal model [14, 6]. The feedback loop between $A\beta$ and Ca^{2+} has also been analyzed

using a deterministic system, adding a noise-inducing element to the system [13, 17]. Models that observe lower-level processes have also been studied, lending to valuable insight as to what is suggested to be seen on the cellular level [11, 9].

It should be noted that in these models, while each has its own way of observing some of the underlying mechanisms, there are still potential shortfalls when it comes to finding effective treatments. Evidence suggests that focusing solely on anti-amyloid beta treatments or tau protein treatments, rather attempting to determine treatments for many different aspects of the disease [1, 15] This provides difficulty in understanding exactly the importance of which component of the disease should be the focus of further study in order to provide evidence for treatments having higher efficacy. In this study, a mechanistic model is proposed, analyzing the interaction between $A\beta$, Ca^{2+} , and tau proteins; observing the effects of this multifaceted interaction on neurodegeneration, and cognitive decline. “Sensitivity analysis is performed in order to better understand the importance of each aspect of the model, and optimal control is used in order to determine the suggested avenues of effective treatment.”

2 Proposed Model

To understand the role calcium may have in the interaction of AD, a model is proposed using five biomarkers found to have interactions within the disease: hallmark biomarkers through amyloid beta and tau, an interactional biomarker through calcium, suggested to have a role in amyloid beta and tau protein accumulation. Also observed are neuronal loss biomarkers, and cognitive decline. The pathology of AD is suggested to start through a feedback loop between amyloid beta and calcium through a dysregulation of calcium; this exacerbates the aggregation of amyloid beta, forming plaques. These plaques then promotes the hyperphosphorylation of tau proteins, leading to neurodegeneration and subsequently cognitive decline.

2.1 Amyloid Beta and Calcium Equations

Because amyloid beta and calcium are suggested to interact in such a way that a dysregulation of calcium causes an acceleration of amyloid beta accumulation [13], this is modeled mathematically as

$$\frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca + \sigma} - k_1 A_\beta - u_1 A_\beta \quad \text{with} \quad A_\beta(T_0) = A_0, \quad (1)$$

where A_0 is the initial condition of amyloid beta at age T_0 , and can vary on a per patient basis [6]. Here, $a(Ca/[Ca + \sigma])$ acts as the interaction of calcium on A_β using a Holling-Type II function.

Similarly, an amyloid beta interaction is observed to have a role in the further dysregulation of calcium, and the equation for calcium is defined as

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \quad \text{with} \quad Ca(T_0) = Ca_0. \quad (2)$$

Here, the accumulation of amyloid beta is positive at a rate b_2 , with a natural removal rate of calcium, k_2 .

2.2 Tau Equation

Many studies have suggested that the pathology of tau is initiated by amyloid beta accumulation, and have modeled this process mathematically [18, 6]. However, it has also been suggested that calcium also has an effect inducing the production of tau proteins [5]. This leads to a definition of the tau equation to be given by

$$\frac{d\tau}{dt} = c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau \quad \text{with} \quad \tau(T_0) = \tau_0. \quad (3)$$

It is noted here that only pathological tau is observed, where other studies have used compartments relating to non-amyloid dependent tau accumulation, or suspected non-Alzheimer pathology (SNAP) [14, 6].

2.3 Neurodegeneration Equation

When tau is in a pathological state, it no longer binds to microtubules as intended; instead, the formation of NFTs occur in the neurons, leading to neurodegeneration [2]. Hence, this equation is defined as

$$\frac{dN}{dt} = d_1 + d_2 \tau - k_4 N. \quad (4)$$

2.4 Cognitive Decline Equation

Both tau and neurodegeneration are evidenced to directly affect cognitive decline [3, 6]. Because of this, the equation for cognitive decline is defined as

$$\frac{dC}{dt} = e_1 + e_2 NR + e_3 \tau - k_5 C. \quad (5)$$

The parameter R represents a patient having the APOE $\epsilon 4$ gene, which has been shown to be a risk factor in Alzheimer's disease (need to find citation of this).

This model is summarized as a system of ordinary system of differential equations with

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca+\sigma} - k_1 A_\beta - u_1 A_\beta \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \\ \frac{d\tau}{dt} = c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau \\ \frac{dN}{dt} = d_1 + d_2 \tau - k_4 N \\ \frac{dC}{dt} = e_1 + e_2 NR + e_3 \tau - k_5 C \end{cases} \quad \text{with} \quad \begin{cases} A_\beta(T_0) = A_0 \\ Ca(T_0) = Ca_0 \\ \tau(T_0) = \tau_0 \\ N(T_0) = N_0 \\ C(T_0) = C_0. \end{cases} \quad (6)$$

3 Analysis

3.1 Preliminary Results

Assuming Equation 6 is subject to initial conditions that are non-negative, it is noted that the system is invariant in the non-negative orthant. Considering the components uncoupled from the rest of the proposed model, $dA_\beta/dt, dCa/dt$ can be shown to be bounded. This is done by first noticing $a_1 + a_2 \frac{Ca}{Ca+\sigma} - r_1 A_\beta \leq a_1 + a_2 - r_1 A_\beta$, since $Ca/(Ca + \sigma) \in [0, 1]$. So,

$a_1 + a_2 - r_1 A_\beta < 0$ when $A_\beta > \frac{a_1 + a_2}{r_1}$. Similarly, using that A_β is bounded, it follows that $\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca \leq b_1 + b_2 M - r_2 Ca$ for $M > A_\beta(t) \forall t \geq 0$. So, $b_1 + b_2 M - r_2 Ca < 0$ when $Ca > \frac{b_1 + b_2 M}{r_2}$. Thus Ca is also bounded. This process, when coupled, results in the rest of the system being bounded.

Lemma 1 (Establishment of Unique Equilibria of Uncoupled System). *There exists a unique equilibria (A_β^*, Ca^*) for the system*

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca + \sigma} - k_1 A_\beta - u_1 A_\beta \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \end{cases} \quad (7)$$

Proof. To establish the existence of equilibria for Equation 7, a solution to the following must be obtained:

$$a_1 + a_2 \frac{Ca}{Ca + \sigma} - k_1 A_\beta - u_1 A_\beta = 0 \quad (8)$$

$$b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca = 0. \quad (9)$$

By solving $\frac{dA_\beta}{dt} = a_1 + \frac{a_2 Ca}{Ca + \sigma} - r_1 A_\beta$ for A_β , we get

$$A_\beta = \frac{a_1 Ca + a_2 Ca + a_1 \sigma}{r_1 (Ca + \sigma)}. \quad (10)$$

We then substitute this value into $\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca$. The resulting expression is,

$$b_1 - r_2 Ca + \frac{b_2 (a_1 Ca + a_2 Ca + a_1 \sigma)}{r_1 (Ca + \sigma)}. \quad (11)$$

Setting (11) equal to 0 and doing further simplification yields the quadratic,

$$-(r_1 r_2) Ca^2 + (a_1 b_2 + a_2 b_2 + b_1 r_1 - r_1 r_2 \sigma) Ca + a_1 b_2 \sigma + b_1 r_1 \sigma = 0. \quad (12)$$

Notice that in (12), the quadratic term is always negative and the constant term is always positive. The linear term can either be positive or negative depending on the choice of parameters. Regardless of the parity of the linear term, there is only one sign change in the expression, so by Descartes' Rule of Signs, there is one positive real root to this quadratic. Thus, there exists a unique positive equilibrium (A_β^*, Ca^*) for the system in Equation 7. $\square$

Lemma 2 (Global Stability of Equilibrium). *The equilibrium (A_β^*, Ca^*) is globally asymptotically stable.*

Proof. Let $\mathbf{F}(A_\beta, Ca) = \langle a_1 + \frac{a_2 Ca}{Ca + \sigma} - r_1 A_\beta, b_1 + b_2 A_\beta - r_2 Ca \rangle$. Then, $\nabla \cdot \mathbf{F} = -r_1 - r_2$. Choose the function $\alpha(A_\beta, Ca) = 1$. So, $\nabla \cdot (\alpha(\cdot) \mathbf{F}(\cdot)) < 0$, therefore by the Bendixson-Dulac Theorem, there is no limit cycle in the system of ODEs. So, the equilibrium point must be globally stable in the non-negative orthant. $\square$

Theorem 1 (Global Stability of Equilibrium of System). *There exists a unique globally asymptotically stable equilibrium $(A_\beta^*, Ca^*, \tau^*, N^*, C^*)$.*

Proof. By Lemma 2, there exists a unique globally asymptotic stable equilibrium (A_β^*, Ca^*) . By solving (3) for τ we get the following equilibrium value:

$$\tau^* = \frac{c_1 + c_2 A_\beta^* + c_3 Ca^*}{k_3 + u_3}. \quad (13)$$

By similar methods we find

$$N^* = \frac{d_1 + d_2 \tau^*}{k_4} \quad (14)$$

and

$$C^* = \frac{e_1 + e_2 N^* R + e_3 \tau^*}{k_5}. \quad (15)$$

Thus there is a unique globally asymptotic stable equilibrium solution for the system in (6). $\square$

3.2 Results and Discussion

4 Parameters

5 Optimal Control

5.1 Objective Function

$$J(u_1, u_2, u_3) = \int_0^K A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 dt. \quad (16)$$

5.2 Admissible Controls

$$u = \{(u_1, u_2, u_3) \in L^\infty(0, K)^3 \mid (u_1, u_2, u_3) \in [0, u_1^{\max}], [0, u_2^{\max}], [0, u_3^{\max}] \forall t \in [0, K]\} \quad (17)$$

5.3 Hamiltonian

$$\begin{aligned} \mathcal{H} = & A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 + \lambda_1 \left(a_1 + a_2 \frac{Ca}{Ca + \sigma} - k_1 A_\beta - u_1 A_\beta \right) \\ & + \lambda_2 (b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca) + \lambda_3 (c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau) \\ & + \lambda_4 (d_1 + d_2 \tau - k_4 N) + \lambda_5 (e_1 + e_2 NR + e_3 \tau - k_5 C). \end{aligned} \quad (18)$$

Var	Value	Unit	Source
A_β	~ 0.01	C	[14]
Ca	0		
τ	0	C	[14]
N	0		
C	0		
a_1	0.25	c/yr	[13]
a_2	2.89	c/yr	[13]
b_1	11	c/yr	[13]
b_2	1000	yr^{-1}	[13]
c_1			
c_2	0.15	yr^{-1}	[6]
c_3			
d_1			
d_2			
e_1	1.67	yr_{-1}	[6]
e_2	3.83	yr_1	[6]
e_3			
k_1	0.35	yr^{-1}	[13]
k_2	500	yr^{-1}	[13]
k_3			
k_4	1.00	C	
k_5	169.48	C	
u_1			[13]
u_2			[13]
u_3			
σ	0.00027	C	[17]

5.4 Adjoint Equations

$$\begin{cases} \frac{d\lambda_1}{dt} = \frac{\partial \mathcal{H}}{\partial A_\beta} = k_1\lambda_1 + u_1\lambda_1 - b_2\lambda_2 - c_2\lambda_3 \\ \frac{d\lambda_2}{dt} = \frac{\partial \mathcal{H}}{\partial Ca} = -\lambda_1 \frac{a_2\sigma}{(Ca+\sigma)^2} + k_2\lambda_2 + u_2\lambda_2 - c_3\lambda_3 \\ \frac{d\lambda_3}{dt} = \frac{\partial \mathcal{H}}{\partial \tau} = k_3\lambda_3 + u_3\lambda_3 - d_2\lambda_4 - e_3\lambda_5 \\ \frac{d\lambda_4}{dt} = \frac{\partial \mathcal{H}}{\partial N} = -A_1 + k_4\lambda_4 - e_2R\lambda_5 \\ \frac{d\lambda_5}{dt} = \frac{\partial \mathcal{H}}{\partial C} = -A_2 + k_5\lambda_5 \end{cases} \quad (19)$$

with transversality conditions

$$\lambda_1(K) = \lambda_2(K) = \lambda_3(K) = \lambda_4(K) = \lambda_5(K) = 0. \quad (20)$$

5.5 Characterizations

$$\begin{cases} \frac{\partial \mathcal{H}}{\partial u_1} = 2A_3u_1 - \lambda_1A_\beta = 0, & u_1^* = \min \left\{ u_1^{\max}, \max \left\{ 0, \frac{\lambda_1A_\beta}{2A_3} \right\} \right\} \\ \frac{\partial \mathcal{H}}{\partial u_2} = 2A_4u_2 - \lambda_2Ca = 0, & u_2^* = \min \left\{ u_2^{\max}, \max \left\{ 0, \frac{\lambda_2Ca}{2A_4} \right\} \right\} \\ \frac{\partial \mathcal{H}}{\partial u_3} = 2A_5u_3 - \lambda_3\tau = 0, & u_3^* = \min \left\{ u_3^{\max}, \max \left\{ 0, \frac{\lambda_3\tau}{2A_5} \right\} \right\} \end{cases} \quad (21)$$

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